

QUANTUM COMPUTING BASICS

INTRODUCTION

Quantum computers leverage quantum mechanical phenomena like superposition and entanglement to perform computations impossible for classical computers.

KEY CONCEPTS

- **Qubits:** Quantum bits that exist in superposition
- **Entanglement:** Quantum correlation between particles
- **Decoherence:** Loss of quantum state

EXAMPLE CODE

```
from qiskit import QuantumCircuit

qc = QuantumCircuit(2, 2)
qc.h(0) # Hadamard gate
qc.cx(0, 1) # CNOT gate
qc.measure_all()
```